

Exercises on variable change

Practice for the exam on June 1, 2026

Review: Variable change in higher dimension for the Lebesgue measure

Recall the theorems around change of variables in $\mathbb{R}^d$:

Theorem 1 (Change of variables in $\mathbb{R}^d$). *Let D_1, D_2 be open subsets of $\mathbb{R}^d$, $\phi : D_1 \rightarrow D_2$ a $\mathcal{C}^1$ -diffeomorphism and $f : D_2 \rightarrow \mathbb{R}$ a function. The following assertions are true:*

- a) *f is positive and measurable on D_2 if and only if $(f \circ \phi)|\det J_\phi|$ is positive and measurable on D_1 ;*
- b) *f is integrable on D_2 if and only if $(f \circ \phi)|\det J_\phi|$ is integrable on D_1 .*

where J_ϕ is the Jacobian matrix of ϕ . Moreover, whenever f is positive measurable or integrable, we have:

$$\int_{D_2} f(y) \, dy = \int_{D_1} f(\phi(x)) |\det J_\phi(x)| \, dx$$

This appears cryptic in essence: how do I, as a student, know where to multiply by $|\det J_\phi|$? Or which function ϕ is?

To remember the big idea, it can be good to draw a parallel with single-variable calculus. Typically, if I have the integral $\int f(x) \, dx$ (where $x \in \mathbb{R}^d$) and I wish to apply the change of variables $y = \phi(x) \in \mathbb{R}^d$, then I can take the very simple single-variable $\phi : x \mapsto 2x$ for some function $f : \mathbb{R} \rightarrow \mathbb{R}$. In particular, ϕ is bijective (its reciprocal is $\phi^{-1} : y \mapsto y/2$) and the change of variables $y = \phi(x) = 2x$ gives

$$y = 2x \quad \text{so} \quad \frac{1}{2} \, dy = dx \quad \text{so} \quad \int f(x) \, dx = \int \frac{1}{2} f\left(\frac{y}{2}\right) \, dy$$

Note that here, the coefficient applied is $1/2$, which you can check is equal to $|\det J_{\phi^{-1}}|$. In the same fashion, this example can help to remember:

$$y = \phi(x) \quad \text{so} \quad |\det J_{\phi^{-1}}(y)| \, dy = dx \quad \text{so} \quad \int f(x) \, dx = \int |\det J_{\phi^{-1}}(y)| f(\phi^{-1}(y)) \, dy$$

Note however that **under no circumstance are you to reason this way on your exam paper**. Analogies like this are exclusive to your scrap paper, and only serve to get your ideas straight. Otherwise, you only write the variable change on the exam paper.

To properly execute a change of variables on an exam paper:

- a) Begin by writing out ϕ . Prove that it is bijective, and give its reciprocal if you can express it simply.
- b) Prove that both ϕ and ϕ^{-1} are $\mathcal{C}^1$. You have various methods to do so.
- c) Find out if the variable change you are looking to execute uses ϕ , or ϕ^{-1} . This is when a single-variable analogy can help.

d) Figure out where you are multiplying by $|\det J_\phi|$, and execute the variable change.

Lastly, note that **polar and spherical coordinate changes** require less justification, as you already have access to these results in your coursework. There is no point in writing out the explicit ϕ function or proving that it is a $\mathcal{C}^1$ -diffeomorphism (you will suffer if you attempt to do so).

Exercises

This first exercise focuses on doing a change of variable for single-variable functions. Although not necessary here, it is still good practice to show that ϕ is a $\mathcal{C}^1$ -diffeomorphism.

Question 1. Using two variable changes, show:

$$\int_0^1 \frac{\sin(\log x)}{x \log x} dx = \int_0^{+\infty} \frac{\sin u}{u} du$$

For your knowledge, this integral evaluates to $\pi/2$ (Dirichlet integral).

This next (pseudo-classic) exercise makes you review polar coordinate changes.

Question 2. Using a polar coordinates, after having justified that you may do so, evaluate:

$$I := \int_{-\infty}^{+\infty} e^{-x^2/2} dx = ?$$

HINT - What does I^2 look like (using Fubini)?

Question 3. Let $a, b \in \mathbb{R}$ be fixed, and let $d \geq 1$ be an integer. Define $\phi : x \in \mathbb{R}^d \mapsto a \cdot x + b$ where $a \cdot x$ is scalar multiplication. Show that ϕ is a $\mathcal{C}^1$ -diffeomorphism if and only if $a \neq 0$. What is the determinant of its Jacobian matrix equal to? What if $a = 0$ - is ϕ still differentiable? What is the determinant of its Jacobian matrix equal to then?

Question 4. Using the previous question, show that the one-dimensional Lebesgue measure on $\mathbb{R}$ is translation-invariant, *i.e.*

$$\forall x \in \mathbb{R}, \forall B \in \mathcal{B}(\mathbb{R}), \quad \lambda(x + B) = \lambda(B)$$

where $x + B := \{x + b : b \in B\}$ by definition. Can you use the previous question to find a relation between $\lambda(aB)$, a , and $\lambda(B)$?

Question 5. Using a variable change, show:

$$\int_0^1 x \sqrt{1-x^4} dx = \int_0^{\pi/2} \frac{1}{2} \sin(\theta)^2 d\theta$$

Question 6. Show that the following ϕ is a $\mathcal{C}^1$ -diffeomorphism for the right domain and co-domain.

$$\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}, \quad \phi(x, y) := \left(x^2 + y^2, \frac{x}{\sqrt{x^2 + y^2}} \right)$$

Note that multiple answers can be obtained here (for the domain and co-domain on which ϕ is a $\mathcal{C}^1$ -diffeomorphism).

As the next exercise uses this function again, we will reveal the answer. Both ϕ_1 and ϕ_2 (defined by the above expression) are $\mathcal{C}^1$ -diffeomorphisms.

$$\phi_1 : \mathbb{R} \times (0, +\infty) \rightarrow (0, +\infty) \times (-1, 1)$$

$$\phi_2 : \mathbb{R} \times (-\infty, 0) \rightarrow (0, +\infty) \times (-1, 1)$$

where $(-1, 1)$ is the open segment from -1 to 1 .

Question 7. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be a positive and measurable function. Using the previous question, prove:

$$\int_{\mathbb{R}^2} (g \circ \phi)(x, y) e^{-(x^2+y^2)/2} dx dy = \int_{(0,+\infty) \times (-1,1)} g(u, v) \frac{e^{-u}}{\sqrt{1-v^2}} du dv$$

HINT - Recall that $\lambda^{\otimes 2}(\mathbb{R} \times \{0\}) = 0$. You may write $\mathbb{R}^2 = \mathbb{R} \times (0, +\infty) \cup \mathbb{R} \times (-\infty, 0) \cup \mathbb{R} \times \{0\}$.

Question 8. Evaluate, using a change of variables, the integral:

$$\int_0^{+\infty} \frac{\arctan(y)}{1+y^2} dy = ?$$

HINT - You should get $\pi^2/8$.

END.